Decent Coaching Classes
VIII SB
Academic Conquest - VII

Subject: General Science (Ch. 5 to 6) **Marks: 30**

Q.1 Choose the correct alternative and rewrite the sentence. **[5]**

1. The number of neutrons in a carbon atom with atomic number 6 and mass number 12 is _____.
   (a) 4 (b) 6 (c) 12 (d) 18
2. The maximum number of electrons that the L shell can hold is _____.
   (a) 2 (b) 8 (c) 18 (d) 32
3. The valency of an element with electronic configuration 2, 8, 7 is _____.
   (a) 1 (b) 3 (c) 7 (d) 8
4. Which of the following is a homogeneous mixture?
   (a) Sand and water (b) Salt solution (c) Saw dust and water (d) Iron filings and sulphur
5. The intermolecular force of attraction is strongest in the _____ state.
   (a) solid (b) liquid (c) gaseous (d) all states equally

Q.2 Fill in the blanks using the words given in the bracket. **[4]**

(a) The number of protons in the nucleus of an atom is its _____ number.
(b) Atoms of the same element having different mass numbers are called _____.
(c) The outermost shell of an atom is called the _____ shell.
(d) Matter whose molecules are made of two or more types of atoms joined by chemical bonds is a _____.

Q.3 (A) State whether the following statements are true or false. Correct the false statement(s). [3]

(a) Almost the entire mass of an atom is concentrated in its nucleus.
(b) A neutron carries a positive charge.
(c) The proportion of the constituents in a mixture is always constant.

(B) Match the pairs. [3]

Group 'A'	**Group 'B'**
1. Plum pudding model	a. Rutherford
2. Nuclear model	b. Niels Bohr
3. Stable orbit model	c. J. J. Thomson

Q.4 Give scientific reasons. (Any 2 of 3) **[4]**

(a) Helium and neon are chemically inert.
(b) Hydrogen is combustible and oxygen helps combustion, but water helps to extinguish fire.
(c) A solid has a definite shape and a definite volume.

Q.5 Answer the following questions. (Any 3 of 4) **[6]**

(a) The atomic number of magnesium is 12 and its mass number is 24. Find the number of protons, neutrons and electrons in it and write its electronic configuration.
(b) Write any two uses of isotopes.
(c) Distinguish between a solution and a suspension (two points).
(d) Write the molecular formula of the compound formed by nitrogen (valency 3) and hydrogen (valency 1) using the cross multiplication method.

Q.6 Answer in detail. (Any 1 of 2) **[5]**

(a) Explain Bohr's stable orbit atomic model with its postulates. Draw a neat labelled diagram.
(b) Define element, compound and mixture with one example of each. Write any two points of difference between a compound and a mixture.